

$$T(n) \leq T(n/5) + T\left(\frac{7.2n}{10}\right) + \boxed{cn} >$$

$\swarrow \quad \searrow \quad \swarrow \quad \searrow$
 $\geq 0 \quad \geq 0$

$\left(\frac{1}{5}\right)^1 \left(\frac{7.2}{10}\right)^2$

Claim: $T(n) \leq dn$ for some constant d .
proof: By induction.

choose d large enough $\geq \frac{c}{0.08}$

I.H. Claim is true for 1, ..., n-1

$$T(n) \leq T(n/5) + T\left(\frac{7.2n}{10}\right) + cn$$

$$\leq \boxed{d \cdot n/5 + d \cdot \frac{7.2n}{10} + cn \leq dn}$$

$$0.92 \ln + cn \leq 2n.$$

$$\begin{array}{l} \equiv \\ \equiv \end{array} \quad cn \leq 0.08 n$$

$$\boxed{d \geq \frac{c}{0.08}}$$

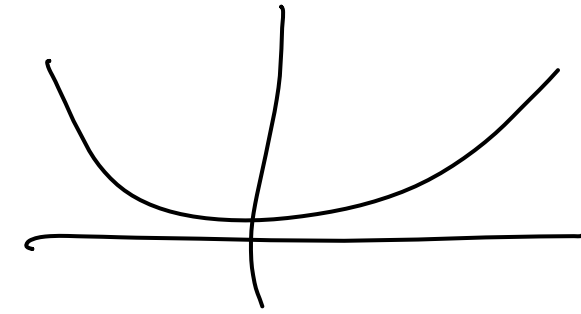
Therefore, $T(n) \leq 2n.$ □
 [Induction step holds]

$$T(n) = O(n), \quad T(n) = \Omega(n)$$

$$T(n) = \Theta(n).$$

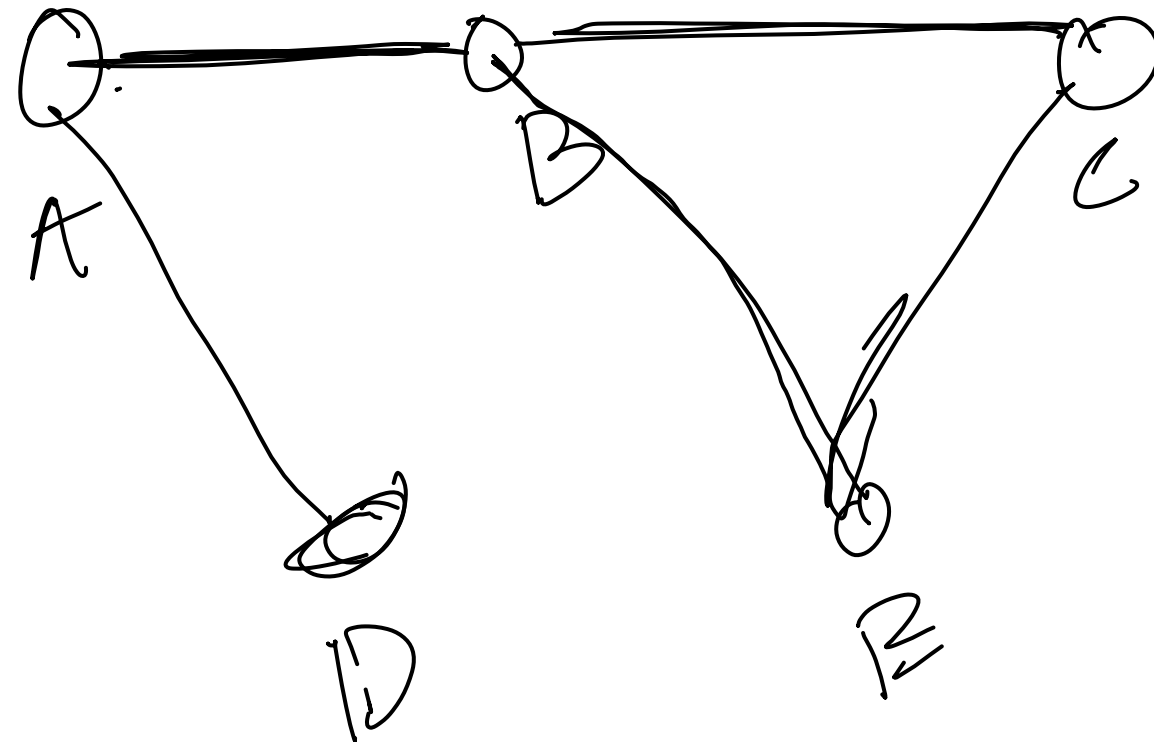
Module 9: Graph Algorithm

(object, relationship)



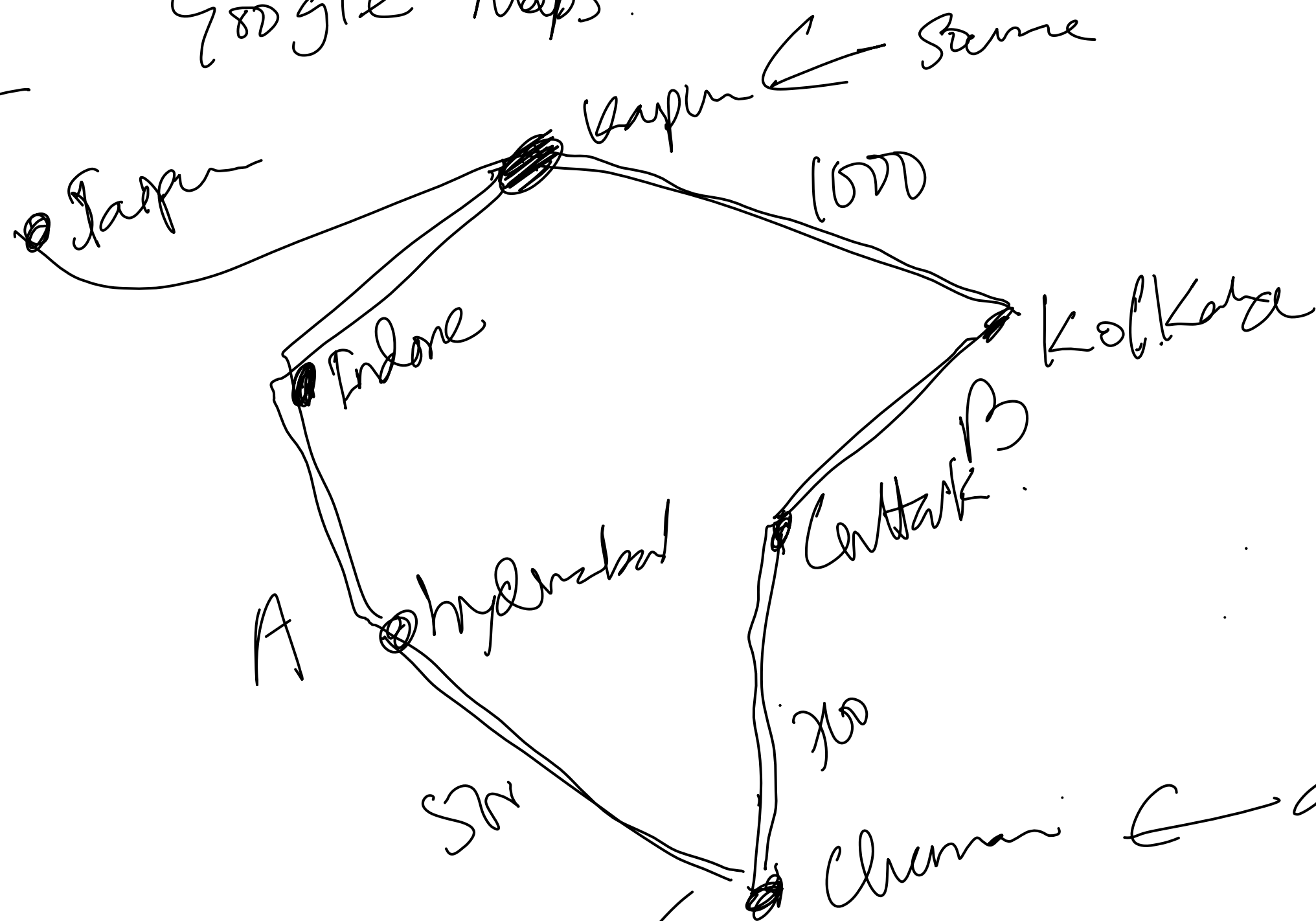
Examples: Facebook graph.

objects: users
relationship: friendship.



Example

Google maps.

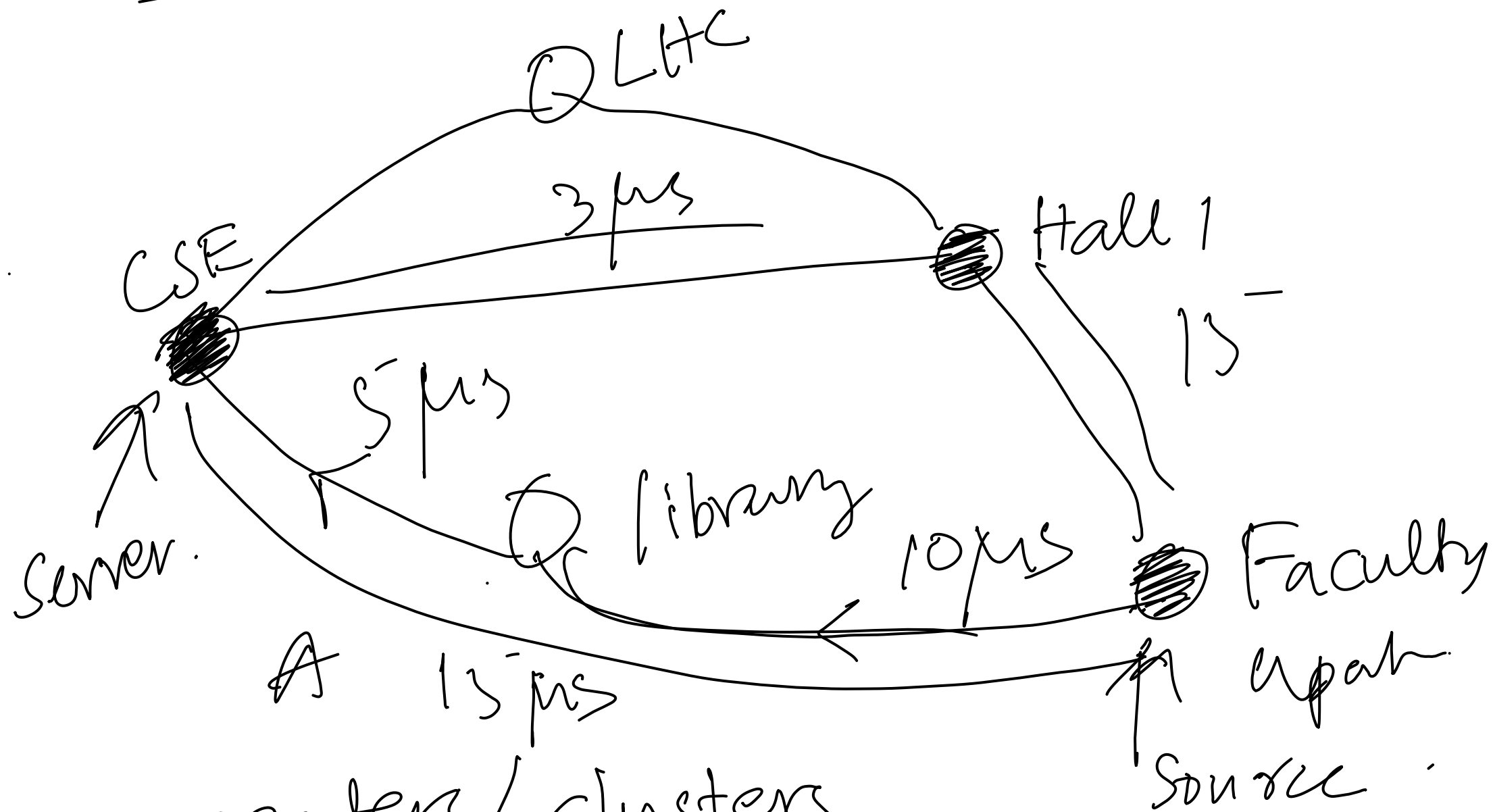


destination

Obj-ct : likes
relationship : road.

Example

Routing.



nodes: routers/clusters

relationship: μ s it-takes to send a packet.

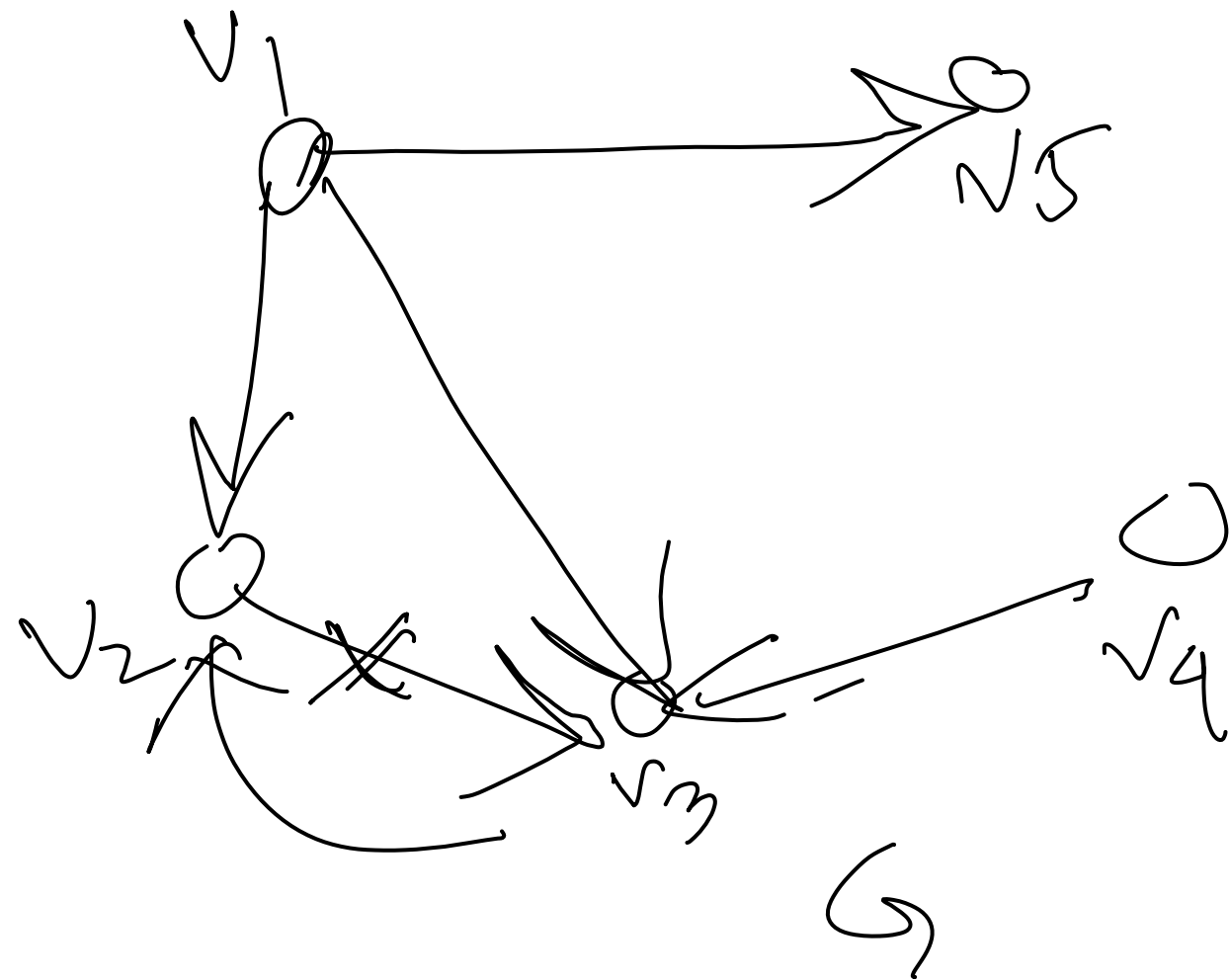
graphs $\equiv$ undirected graphs.

directed graphs -

$A \rightarrow B$

B follows A

$(A, B) \equiv (B, A)$



$V = \{v_1, v_2, \dots, v_5\}$

$E \subseteq V \times V$

$= \{(v_1, v_2), (v_2, v_3), (v_1, v_3), (v_4, v_3), (v_1, v_5)\}$

How are graphs stored inside a computer?

adjacency list

Adjacency matrix

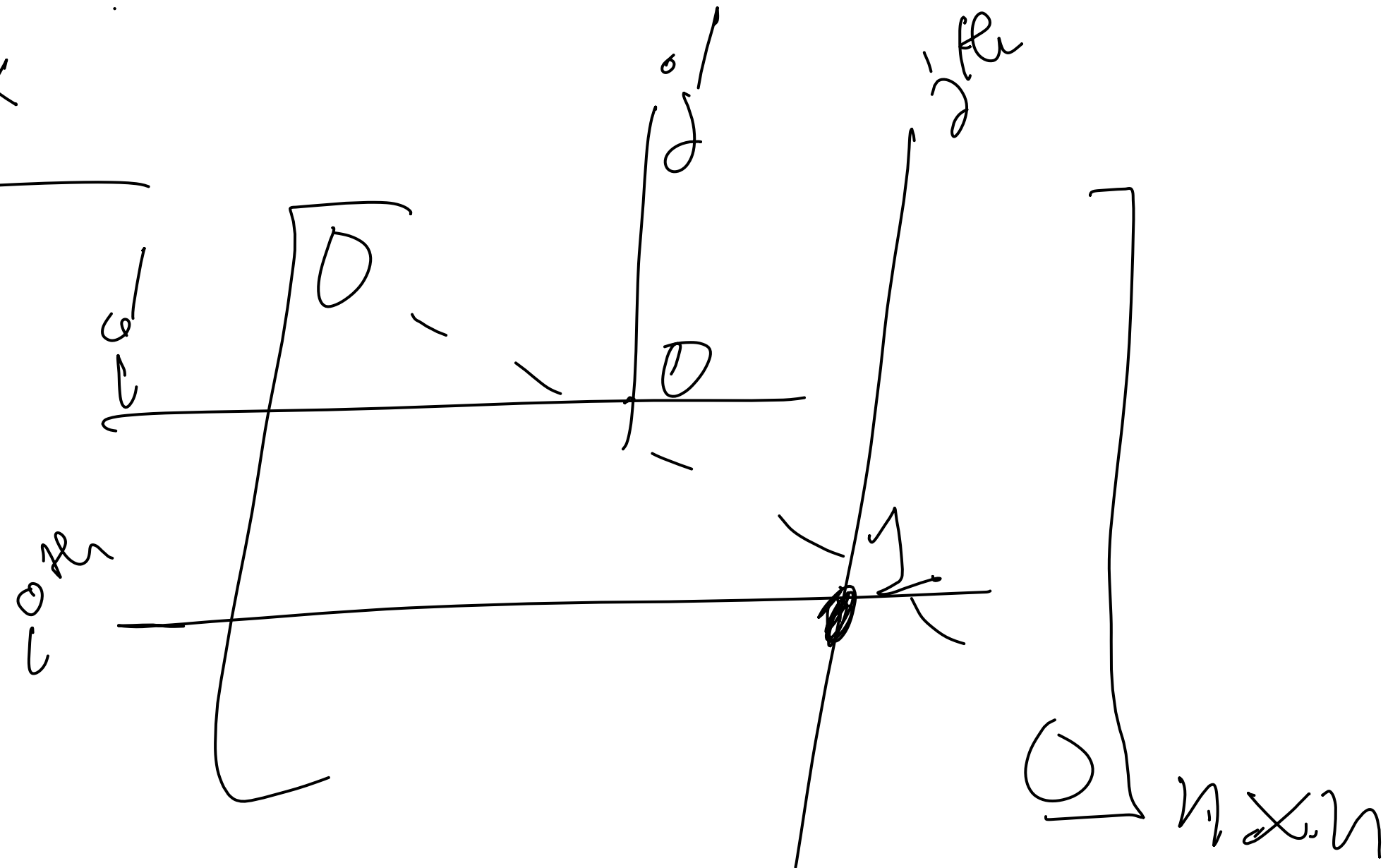
Adjacency matrix

n nodes:

$(i, j) \notin E$

$(i, j) \in E$

row column



	1	2	3	4	5	6
1	0	1	0	0	0	1
2	1	0	1	0	0	0
3	0	1	0	1	0	0
4	0	0	1	0	1	0
5	0	0	0	1	0	1
6	1	0	0	0	1	0



- Symmetric

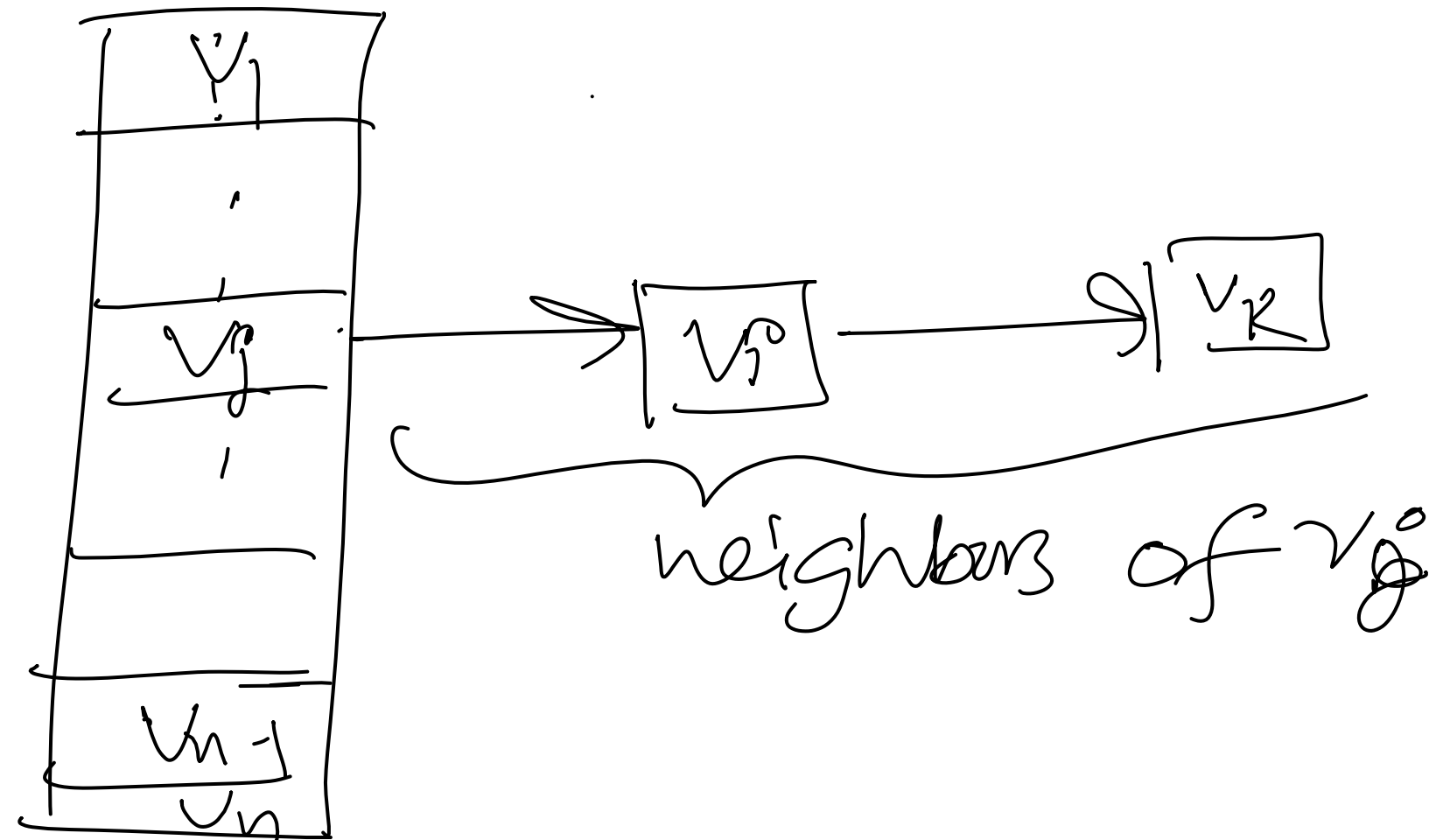
- stored like a directed graph
where each edge is bidirected.

$$\begin{array}{c}
 1 \\
 2 \\
 3 \\
 4 \\
 5
 \end{array}
 \begin{bmatrix}
 1 & 2 & 3 & 4 & 5 \\
 0 & 1 & 1 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 0 & 0 & 0 \\
 0 & 0 & 1 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0
 \end{bmatrix}$$

no longer symmetric.

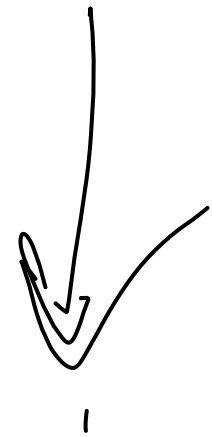
Remark: We'll focus on designing algorithms for directed graphs. More often, they will extend to undirected graphs.

Adjacency list



(v_j, v_R)
 (v_j, v_R)

$$T(m) \leq T(m/5) + T\left(\frac{7g}{2} + 3\right) + cn$$



$$g = m/5 \quad ? \quad \frac{7g}{2} = \frac{7 \cdot m}{10}$$

$$m \leq n$$

$$\underline{\underline{T(n)}} \leq T(n/5) + T\left(\frac{7n}{10} + 3\right) + \underline{\underline{cn}}$$

$$\leq \frac{7.2n}{10}$$

for n large enough.